Protein Modeling

From Homology Modeling to AlphaFold

Christine Schulz
Bio-Structure Hub @ SSC, IWR

Protein Modeling

What do you want to model?

Protein Structures: Full 3D protein structures

Mutations & Changes:

- Single-point mutations
- engineered loops
- Unresolved loops/chain termini
- Missing atoms (especially hydrogens)

Interactions & Modifications:

- Ligand interactions (covalent and non-covalent)
- Metal ions, Cofactors, Substrates, and Inhibitors
- Post-translational modifications
- Macromolecules and Water
- Oligomers

Key Question: What do I want to use my protein model(s) (experimental or predicted) for? (This defines what effect errors in the models will have)

Proteins: Structure Dictates Function

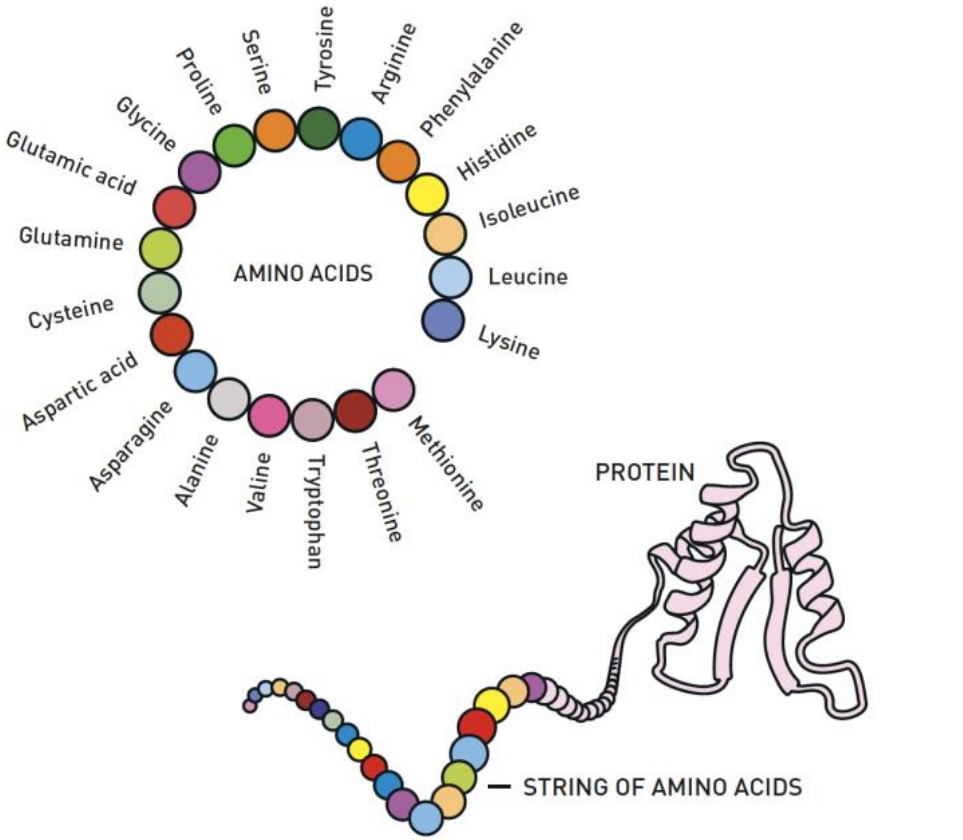
Why do we need 3D Structures?

The Foundation: A protein begins as a simple string of amino acids linked by peptide bonds

The Folding Process: This string coils and folds onto itself, into a distinct structure, driven by internal forces: Electrostatic forces, hydrogen bonds, and solvent interactions

The Significance of Shape: The final **3D structure** is what truly matters:

- dictates function and purpose of the protein through interactions (e.g., hemoglobin having the perfect binding site for oxygen)



©Johan Jarnestad/The Royal Swedish Academy of Sciences

Structure Determination

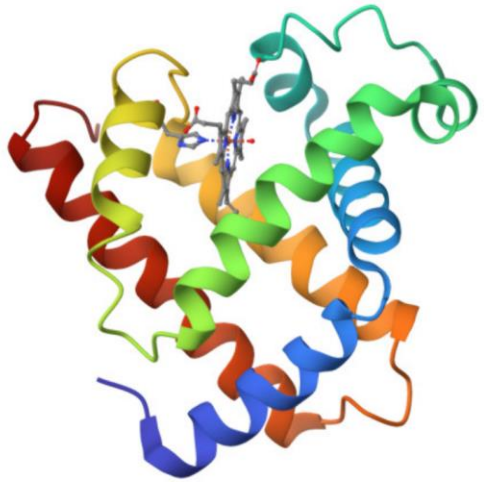
X-Ray Crystallography

Create a crystal of the protein, exposing it to X-rays to generate a diffraction pattern, and then working backward to deduce the molecular shape

The First Success: British biochemist John Kendrew took **12 years** to solve the structure of myoglobin and was awarded with the 1962 Nobel Prize in Chemistry

The Challenge:
The process is difficult and expensive, sometimes costing **tens of thousands of dollars per protein**

https://www.nobelprize.org/prizes/chemistry/1962/summary/



1MBN | pdb_00001mbn

The stereochemistry of the protein myoglobin

Watson, H.C., Kendrew, J.C.

(1969) Prog Stereochem **4**: 299

Released	1976-05-19
Method	X-RAY DIFFRACTION 2 Å
Organisms	Physeter macrocephalus
Macromolecule	MYOGLOBIN (protein)
Unique Ligands	HEM, OH

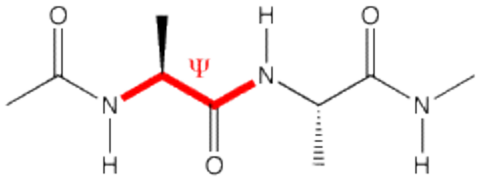
Levinthal's Paradox

A thought experiment

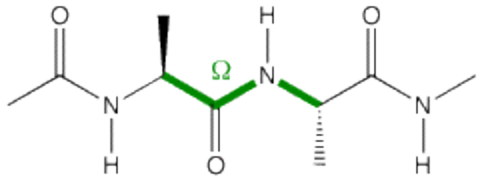
3 angles per amino acid:
phi Φ, psi Ψ, omega Ω



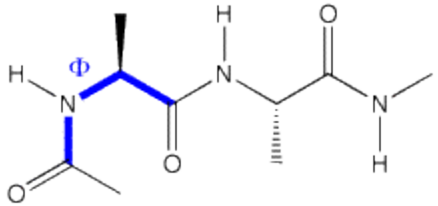
phi (ala1) = 180.0



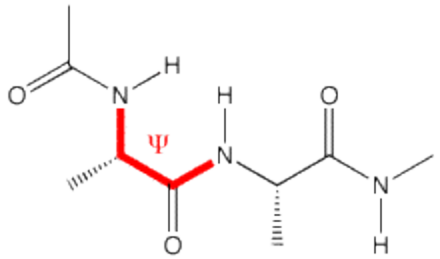
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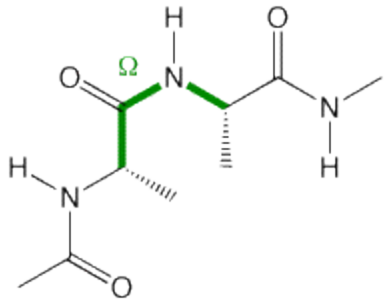
omega (ala1) = 180.0



phi (ala1) = 0.0



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https://www.cup.uni-muenchen.de/ch/compchem/tink/tink2.html

Levinthal's Paradox

A thought experiment



3 angles per amino acid:
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For a 100 AA protein : 3^{100} conformations

Assume a sample speed of 10^{12} conformations /second

Zwanzig R, Szabo A, Bagchi B. Levinthal's paradox. Proc Natl Acad Sci U S A. 1992 Jan 1;89(1):20-2. doi: 10.1073/pnas.89.1.20

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Assume a sample speed of 10^{12} conformations /second

It takes 5.15×10^{35} seconds.
Or 1.63×10^{28} years

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Molecular Modeling

What can we do?

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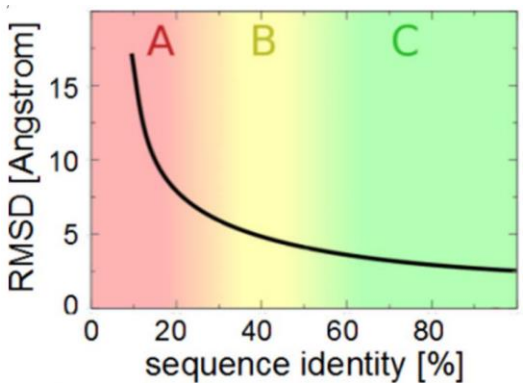
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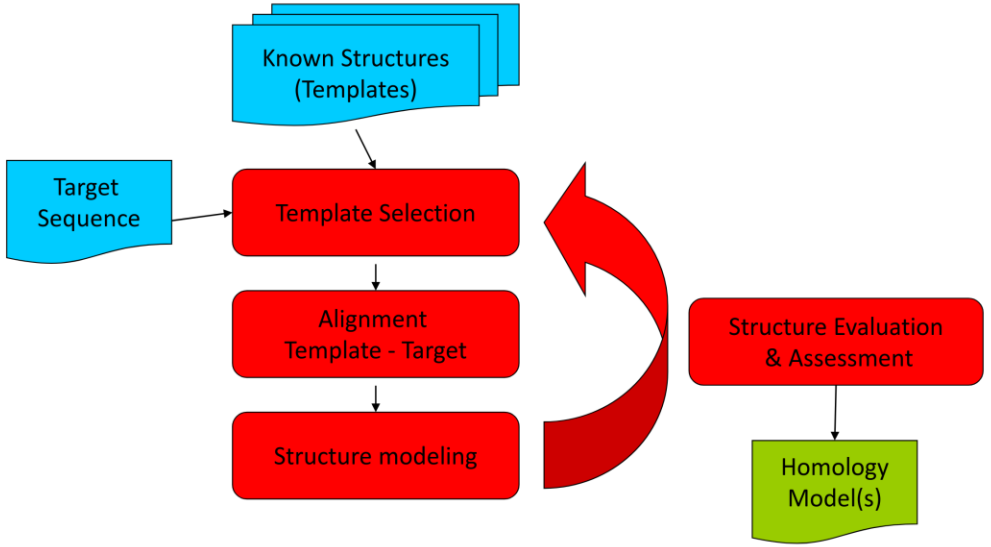
Template based Modeling: Homology Modeling



Idea: target sequence is similar to another sequence with known structure

> 50% sequence identity: Considered a “reliable” model
> 30% sequence identity: Considered a “useful” model
< 30% sequence identity: The resulting model “might be useful”

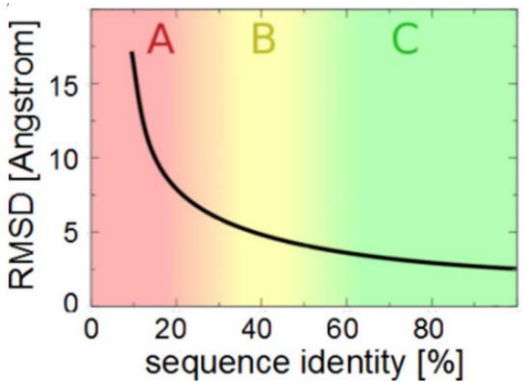
- Multi-sequence alignment (MSA)
- Structure based alignment



https://swissmodel.expasy.org/course

Molecular Modeling

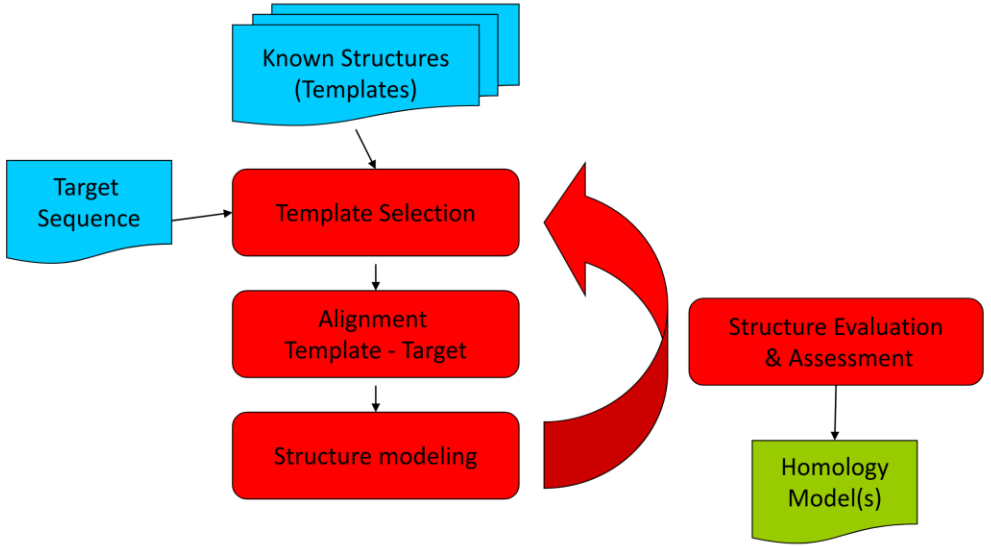
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How to model proteins without templates?
Ab initio modeling

https://swissmodel.expasy.org/course

The Challenge Competition

Critical Assessment of Structure Prediction (CASP)

Started in **1994** to design computer models that could predict structure from sequence:

- Double blind: neither predictors nor organizers and assessors know the structure of the target protein
- Soon to be solved X-ray or NMR structures

- Evaluated by Cα positions
- Perfect agreement: Score of 100
- Target: Score of **90 or above** indicated the structure was solved
- Initial Results: Teams struggled, maximum score 40

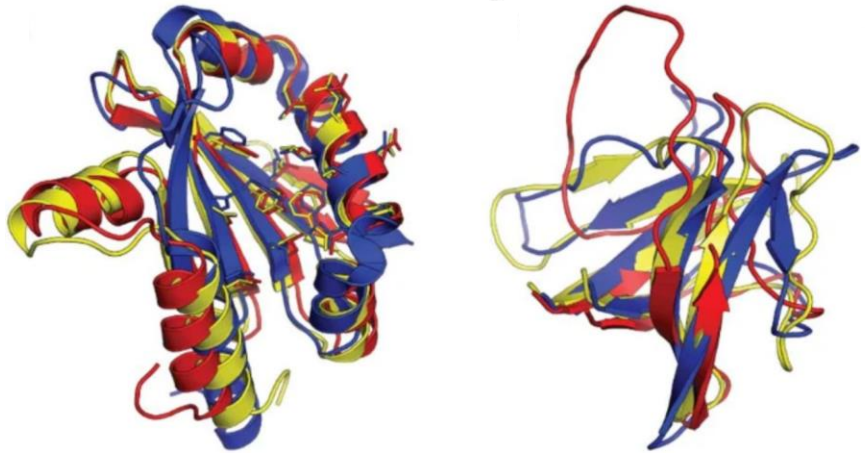
https://predictioncenter.org/index.cgi

Foldit - Fold Protein 3D Structures at Home

Critical Assessment of Structure Prediction (CASP)

Scientific Software Center

- Based on Rosetta (Baker lab)
- Folding protein structures at idle computers (at home!)
- Solved the CASP9 target T0581, an enzyme in HIV



Brief Communication | Published: 18 September 2011

Crystal structure of a monomeric retroviral protease solved by protein folding game players

Firas Khatib, Frank DiMaio, Foldit Contenders Group, Foldit Void Crushers Group, Seth Cooper, Maciej Kazmierczyk, Miroslaw Gilski, Szymon Krzywda, Helena Zabranska, Iva Pichova, James Thompson, Zoran Popović, Mariusz Jaskolski & David Baker

Nature Structural & Molecular Biology **18**, 1175–1177 (2011) | Cite this article

10k Accesses | **421** Citations | **477** Altmetric | Metrics

Meanwhile at DeepMind

AlphaGo

- Go was considered a grand challenge for AI with **10^{170}** possible board configurations
- AlphaGo a combined neural network

October 2015: 5:0 against European Champion Fan Hui
March 2016: 4:1 against 18 world title winner Lee Sodol



https://deepmind.google/research/projects/alphago/

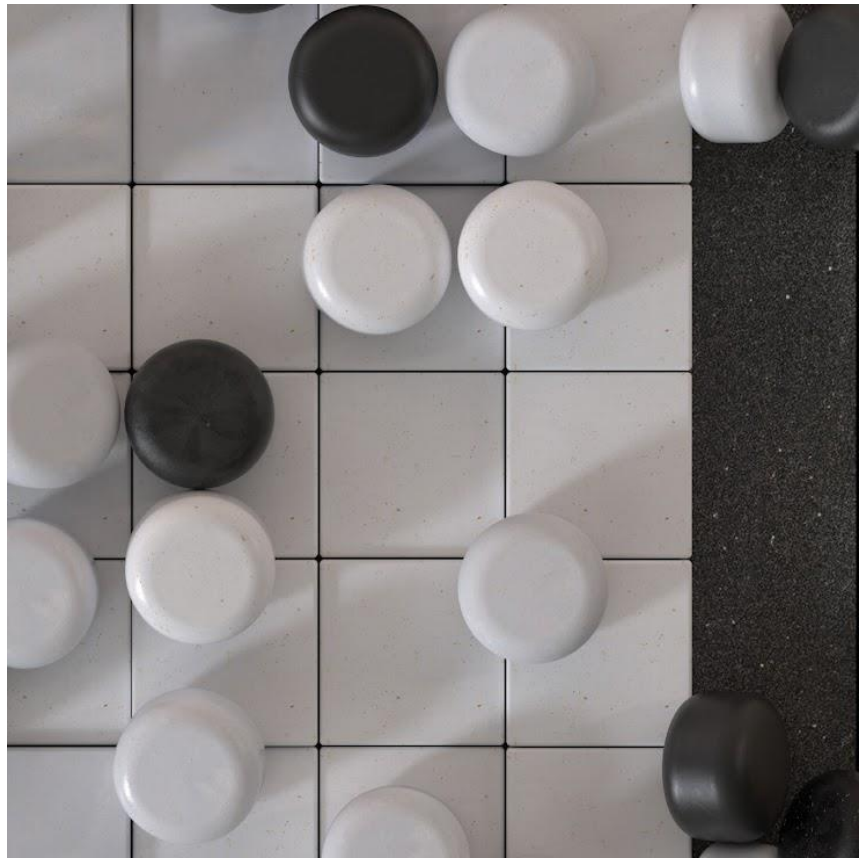
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Could an AI be trained to play Foldit?



https://deepmind.google/research/projects/alphago/

AlphaFold 1in CASP13

Evolutionary clues and a deep neural network

Key Input:

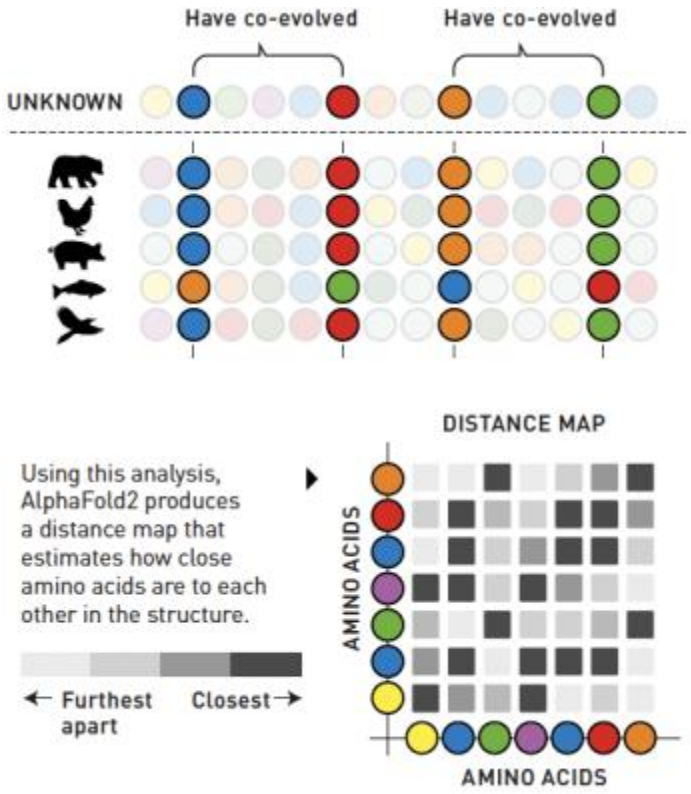
- As species evolve, proteins must retain their functional shape
- AlphaFold used MSAs to look for **co-evolution**: mutations that happen in pairs to retain stability (e.g., swapping attracting charges)

This identified amino acids that must be close to each other in the final structure

Output: Instead of 3D, AF1 predicted a simpler 2D pair representation (distances and torsion constraints)

Result: AlphaFold 1 was the clear winner of CASP 13, but its score of **70** still fell short of the 90-point threshold

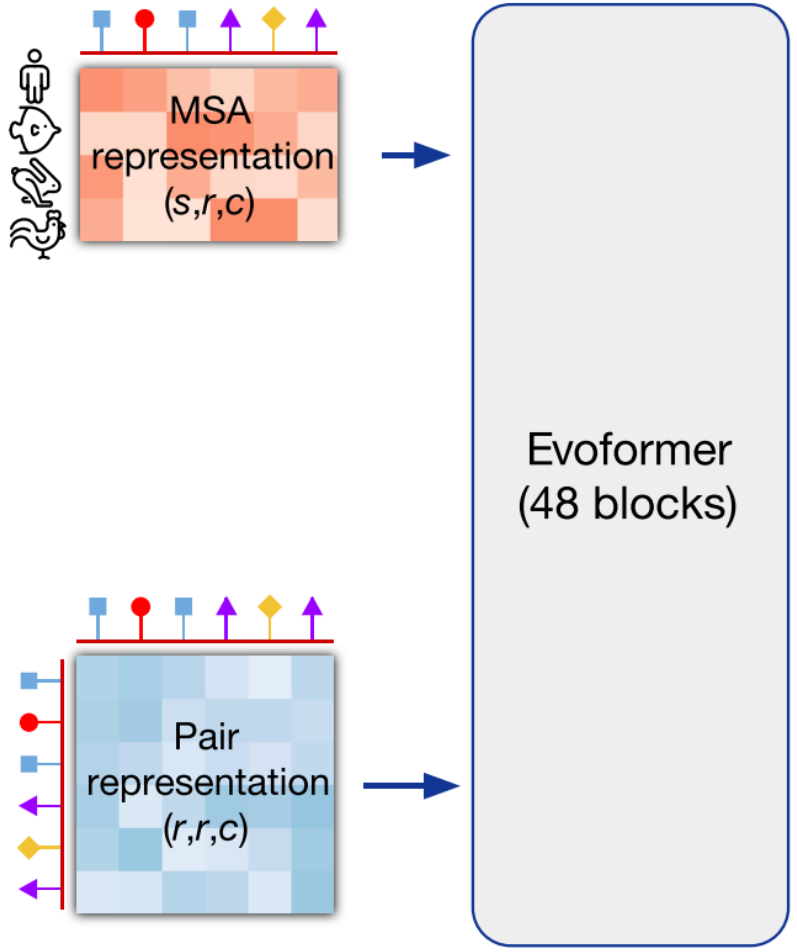
https://www.predictioncenter.org/casp13/index.cgi



AlphaFold 2:
Achieving Experimental Accuracy

The Improvement: AlphaFold 2 achieved its tremendous accuracy boost through **better machine learning algorithms**, not just more data
(it was trained on the *exact same data* as AF1)

- **Innovation 1: The EVO Former**
  - A custom version of the Transformer architecture
  - Uses two parallel areas (Biology and Geometry) that constantly exchange newly found clues



Jumper, J., … Hassabis, D. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature 2021 596:7873*, *596*(7873), 583–589. https://doi.org/10.1038/s41586-021-03819-2

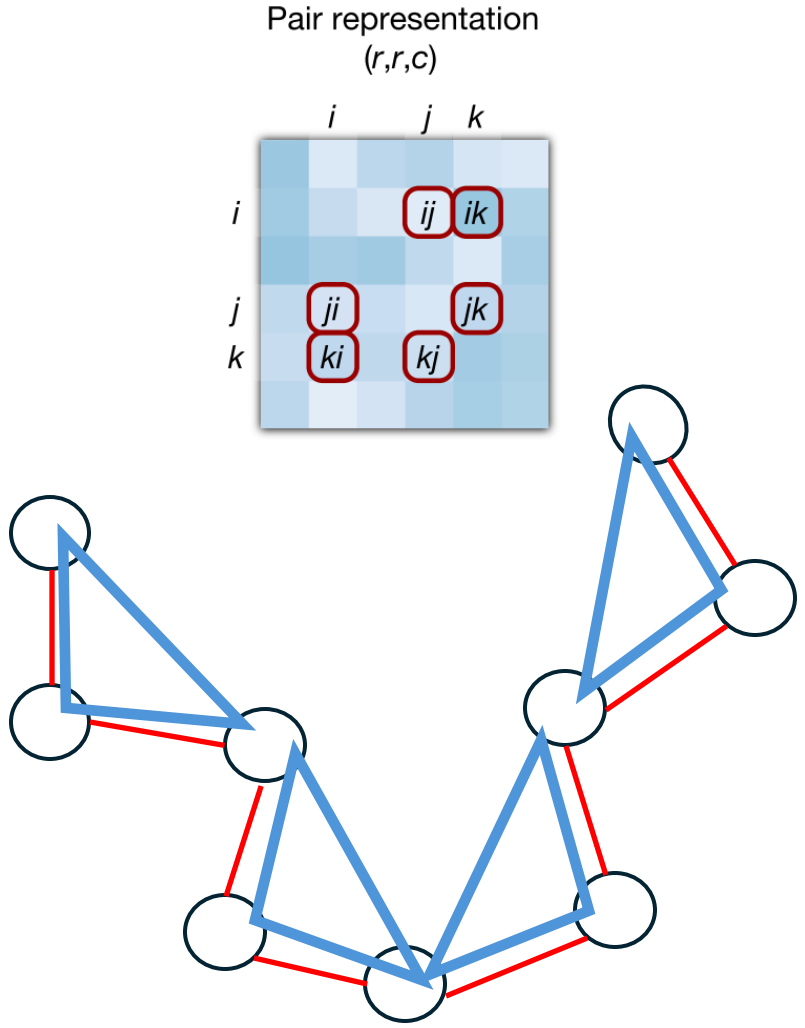
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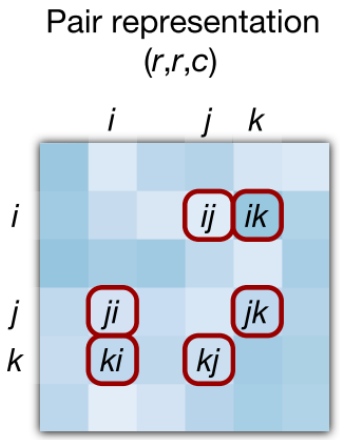
- **Triangular Attention:** Applies the triangle inequality theorem (sum of two sides must be greater than the third) to constrain distances, ensuring a **self-consistent picture** of the structure

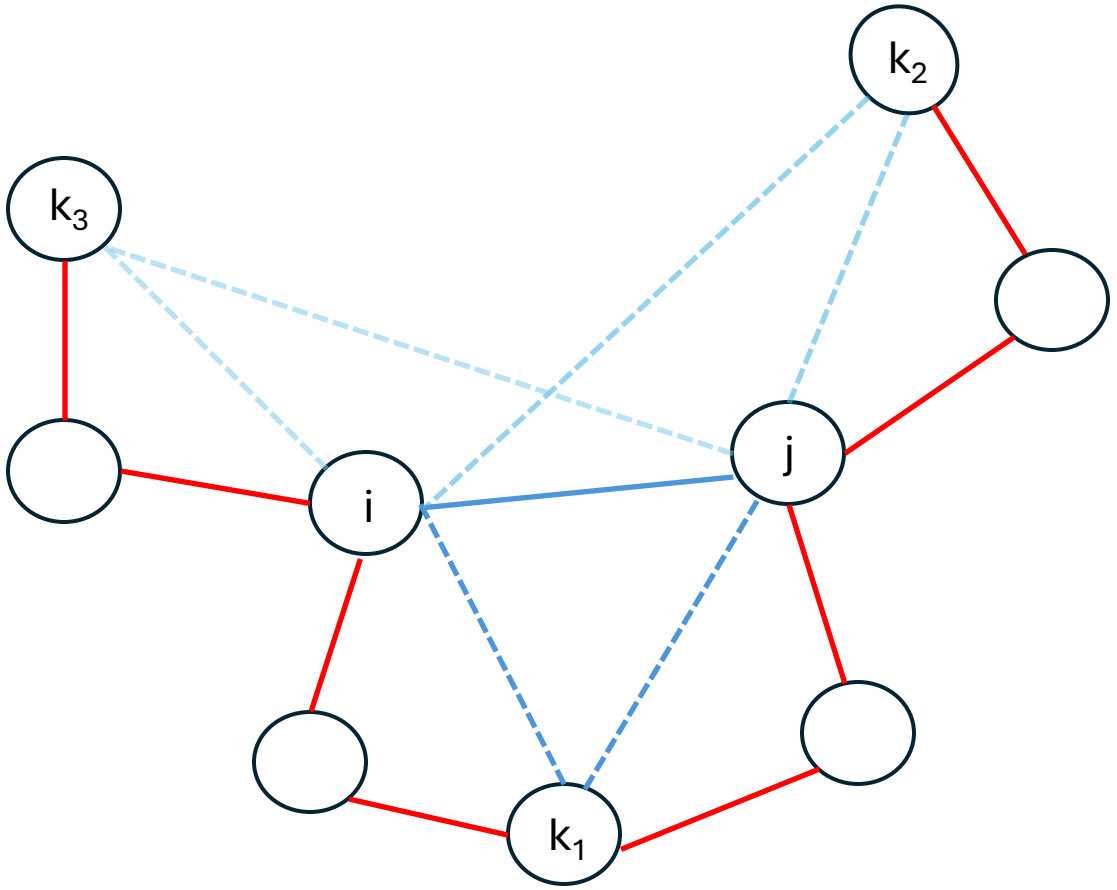


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Triangular Attention

Geometry in Protein 3D Structures



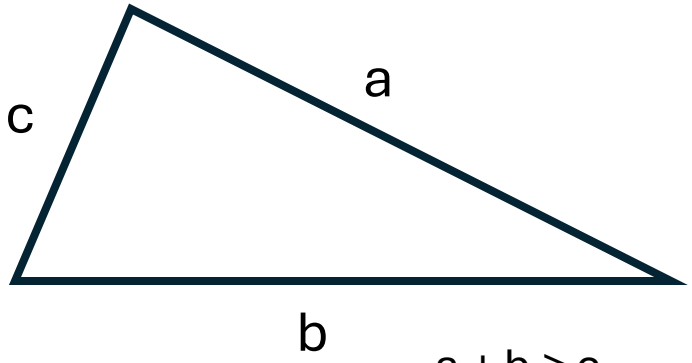


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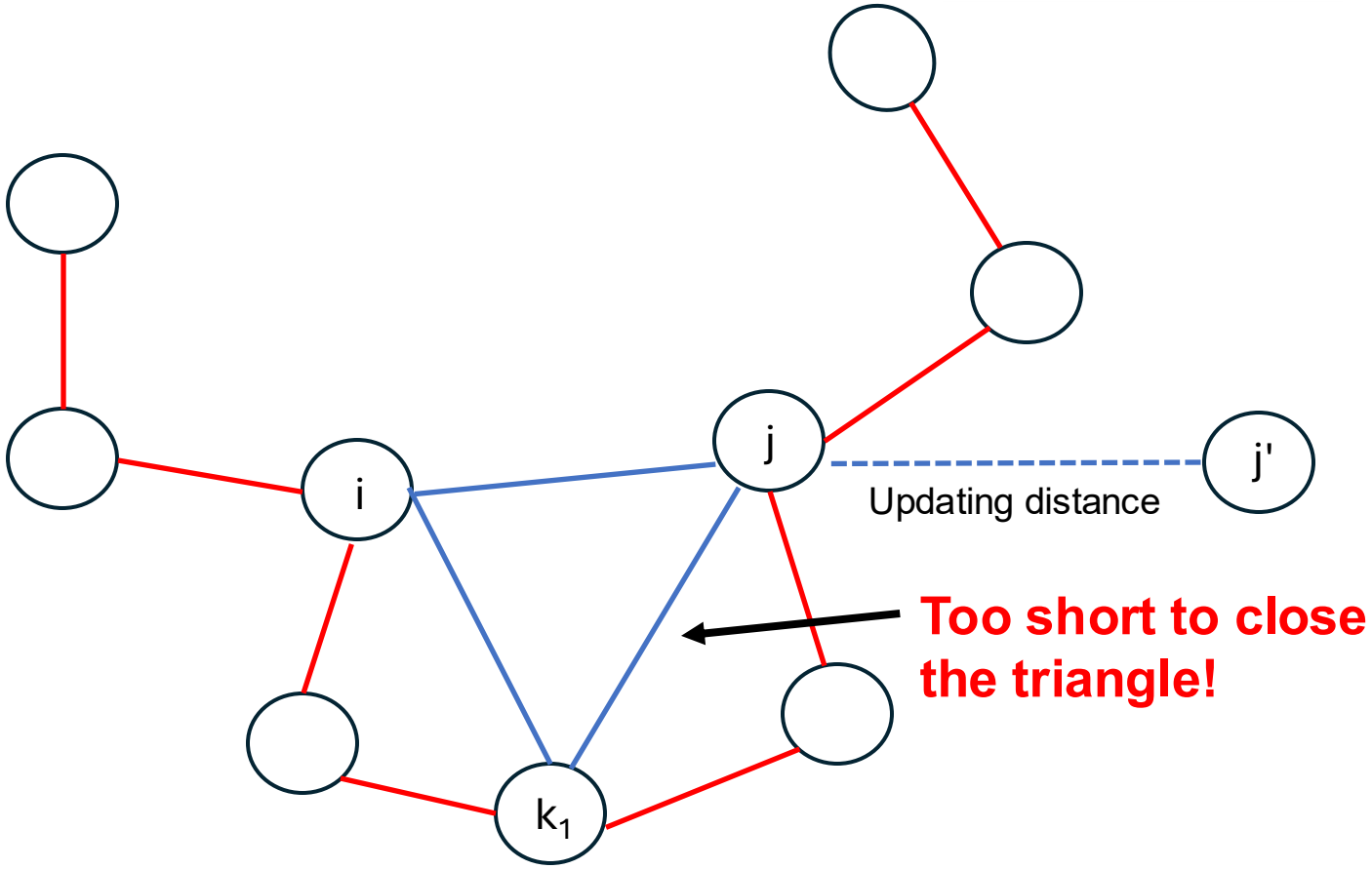
Triangular Attention
Geometry in Protein 3D Structures

Scientific Software Center

Triangle inequality theorem



a + b > c
a + c > b
b + c > a



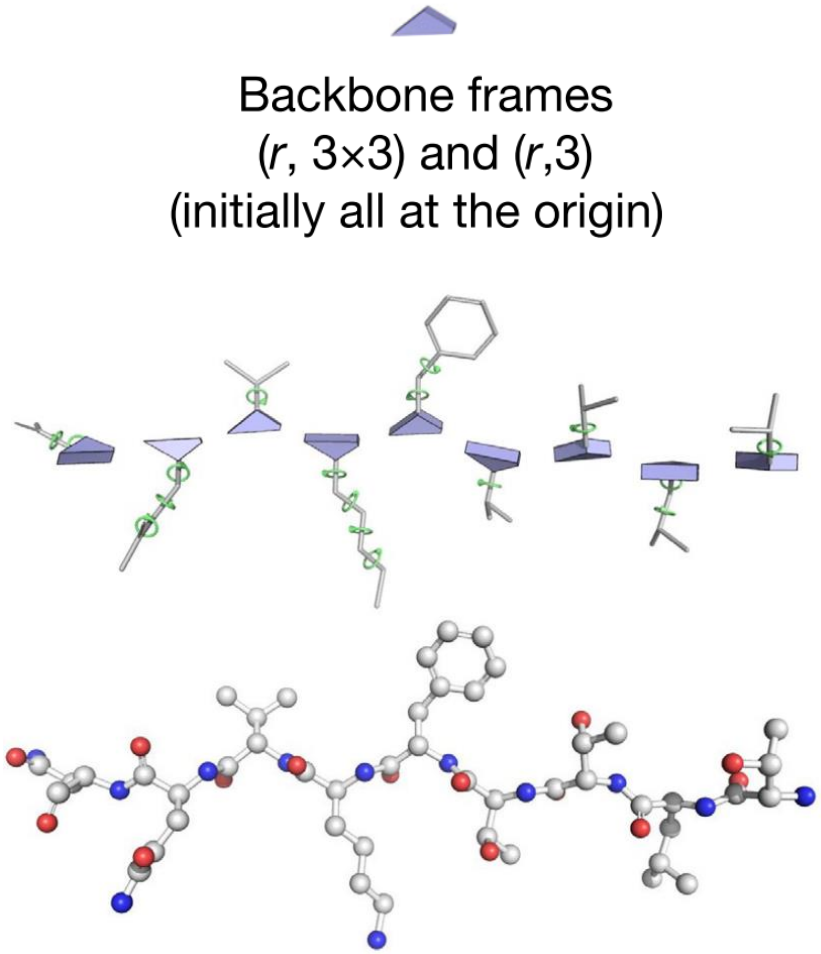
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- **Innovation 1: The EVO Former**
- **Innovation 2: The Structure Module**
  - Takes refined geometric data and predicts the translation and rotation needed to position key atoms

  - Crucially, it treats the input as a "bag of amino acids," allowing the constraint of forming a chain to **emerge naturally** later in the process



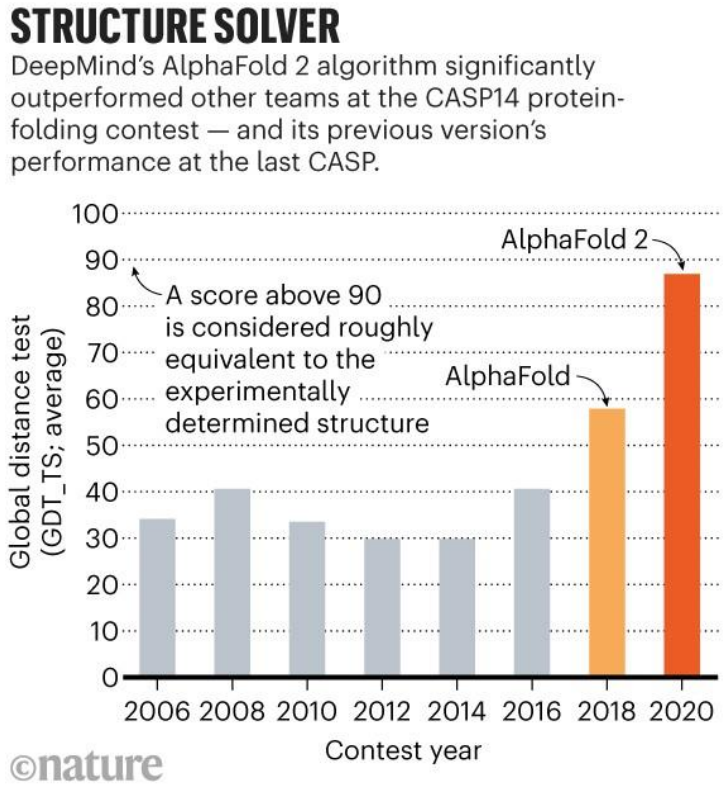
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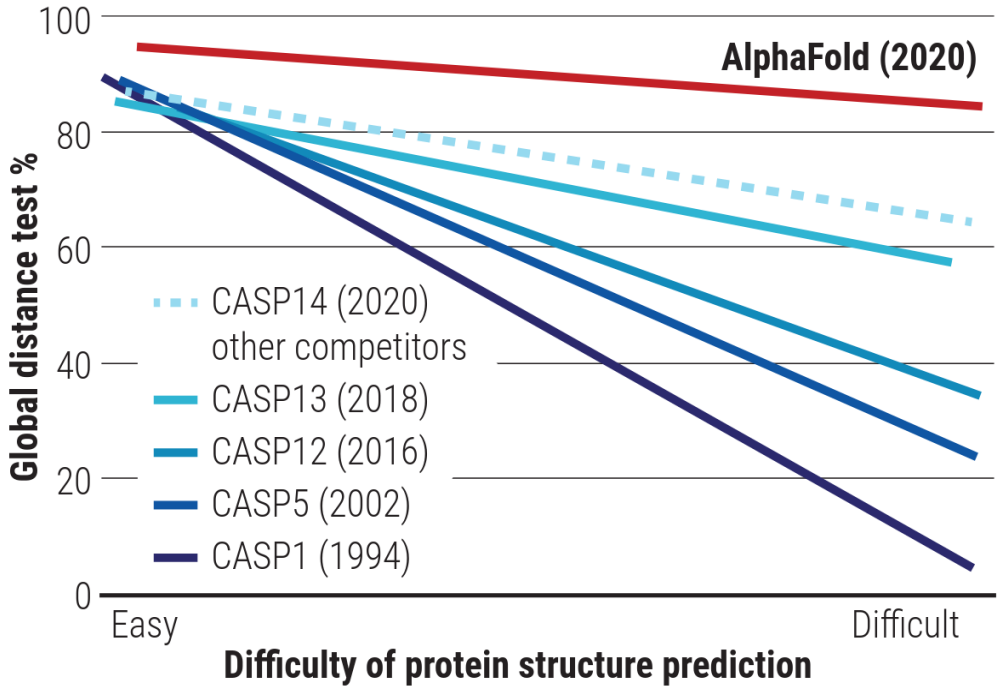
Callaway, E. (2020). "It will change everything": DeepMind's AI makes gigantic leap in solving protein structures. *Nature*, *588*(7837), 203–204. https://doi.org/10.1038/D41586-020-03348-4

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Service, R. (2020). 'The game has changed.' AI triumphs at solving protein structures. *Science*. https://doi.org/10.1126/SCIENCE.ABF9367

AlphaFold 2:
Limitations

Only able to predict **pure protein structures**

- No complexes / protein-protein interactions
- No PTMs
- No lipids or other cofactors
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Minkyung Baek
@minkbaek

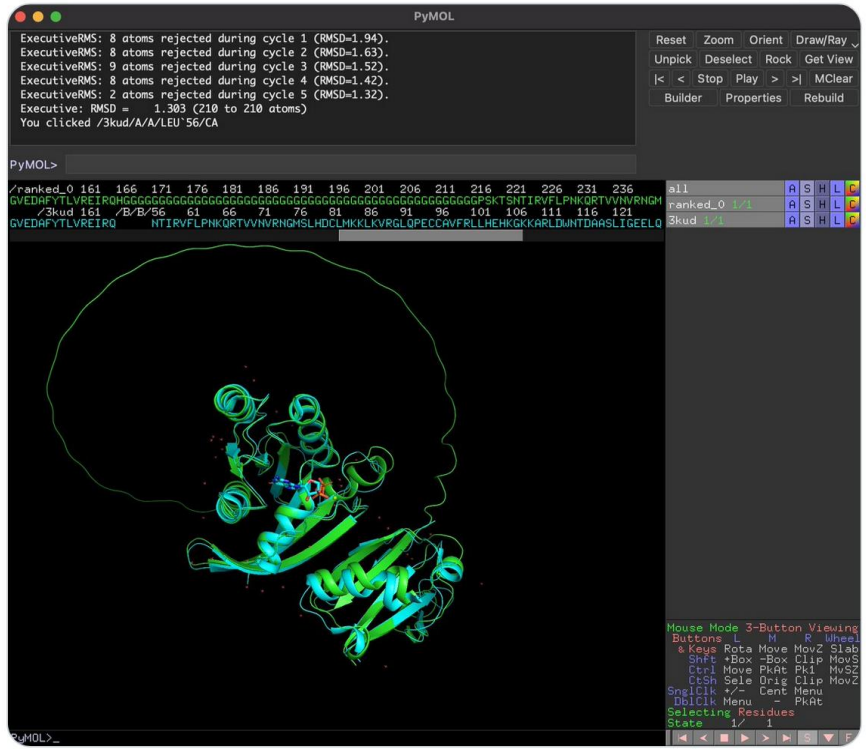
Adding a big enough number for "residue_index" feature is enough to model hetero-complex using AlphaFold (green&cyan: crystal structure / magenta: predicted model w/ residue_index modification).
#AlphaFold #alphafold2

7:33 PM · Jul 20, 2021



Yoshitaka Moriwaki
@Ag_smith

AlphaFold2 can also predict heterocomplexes. All you have to do is input the two sequences you want to predict and connect them with a long linker.



12:07 PM · Jul 19, 2021

Evans, R., … Hassabis, D. (2021). *Protein complex prediction with AlphaFold-Multimer*. https://doi.org/10.1101/2021.10.04.463034

Continuous Evolution: AlphaFold3

Going beyond Protein Structures to Biomolecules

Proteins and Complexes:

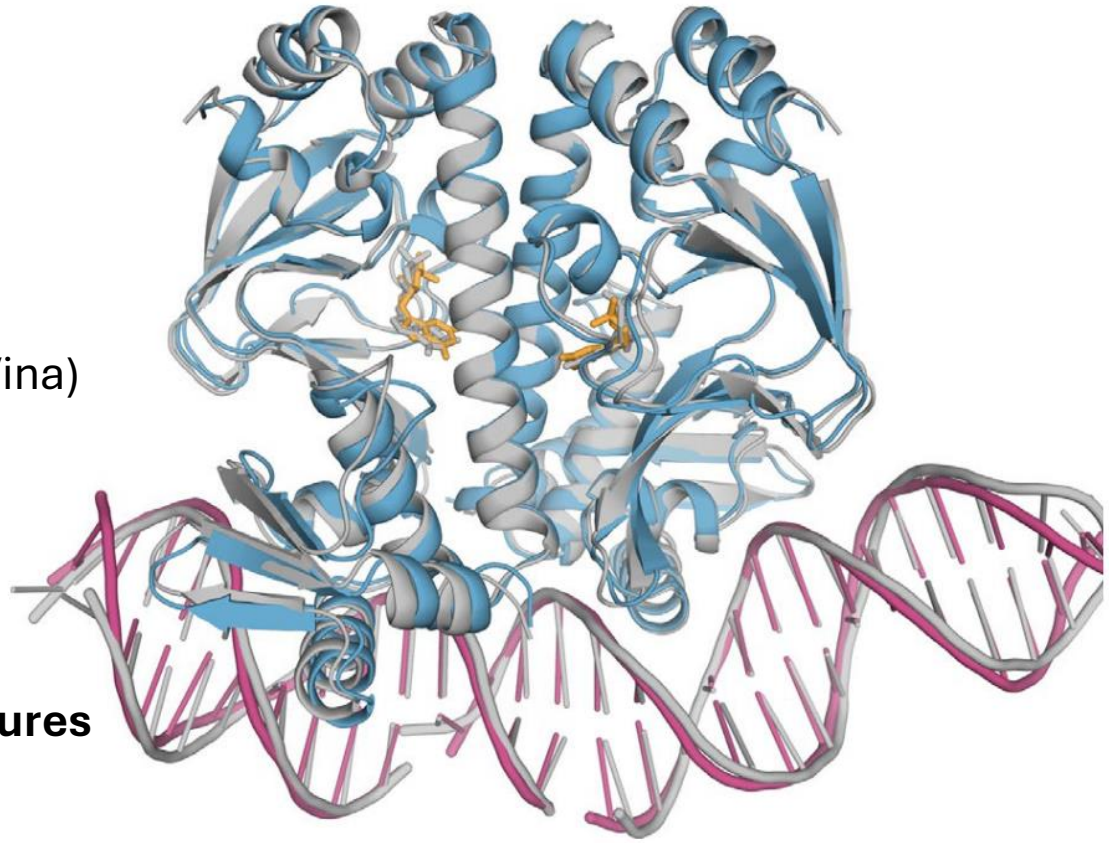
- protein structure and protein–protein interactions
- **antibody–antigen prediction**

Ligands (Small Molecules):

- Demonstrates **protein–ligand interactions** comparable with state-of-the-art docking tools (e.g., Vina)
- Successfully predicts covalent modifications (bonded ligands, glycosylation, modified protein residues and nucleic acid bases)

DNA and RNA Complexes:

- Predicts **protein–nucleic complexes and RNA structures with higher accuracy** than specialized methods like RoseTTAFold2NA



Abramson, J., Adler, J., Dunger, J., … Jumper, J. M. (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, *630*(8016), 493–500. https://doi.org/10.1038/s41586-024-07487-w

Continuous Evolution: AlphaFold3

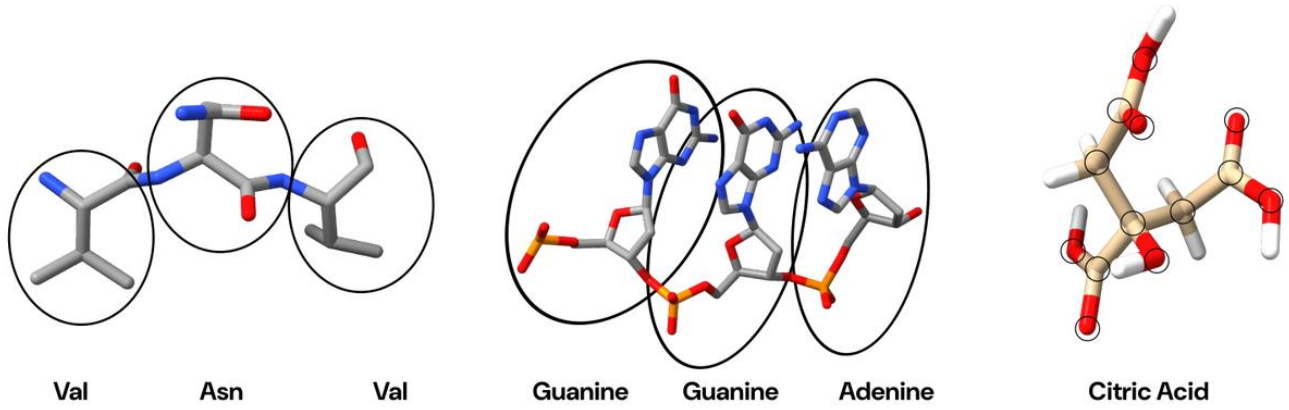
Going beyond Protein Structures to Biomolecules

How is AlphaFold3 able to treat more than amino acids?

- **Tokenization**

AF2: Tokens correspond to amino acids

AF3: Tokens depend on the molecule type



https://www.ebi.ac.uk/training/online/courses/alphafold/

Continuous Evolution: AlphaFold3

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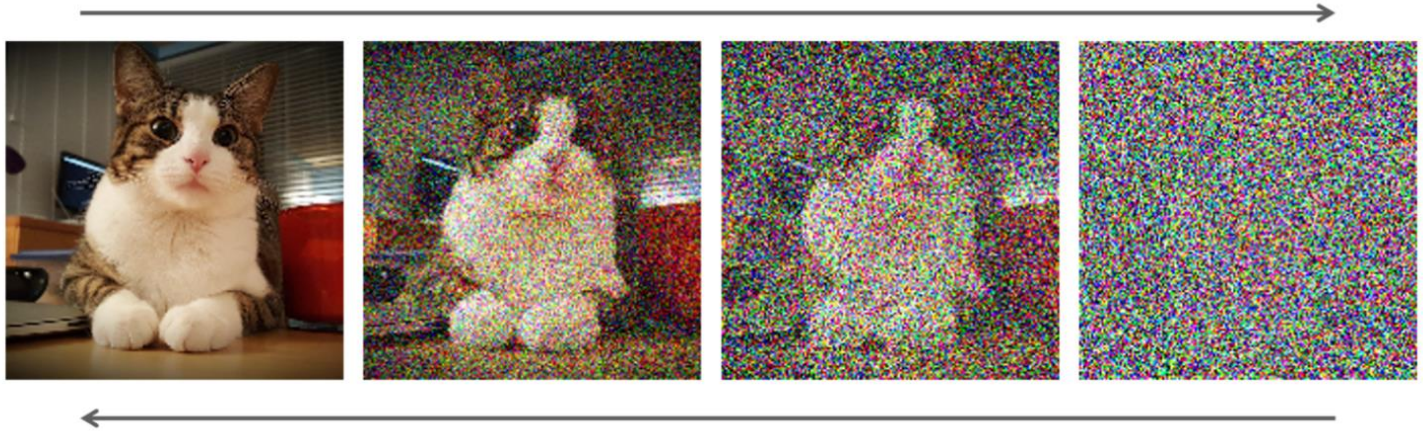
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AF3: Tokens depend on the molecule type

- **Diffusion**

AF2: non-generative AI

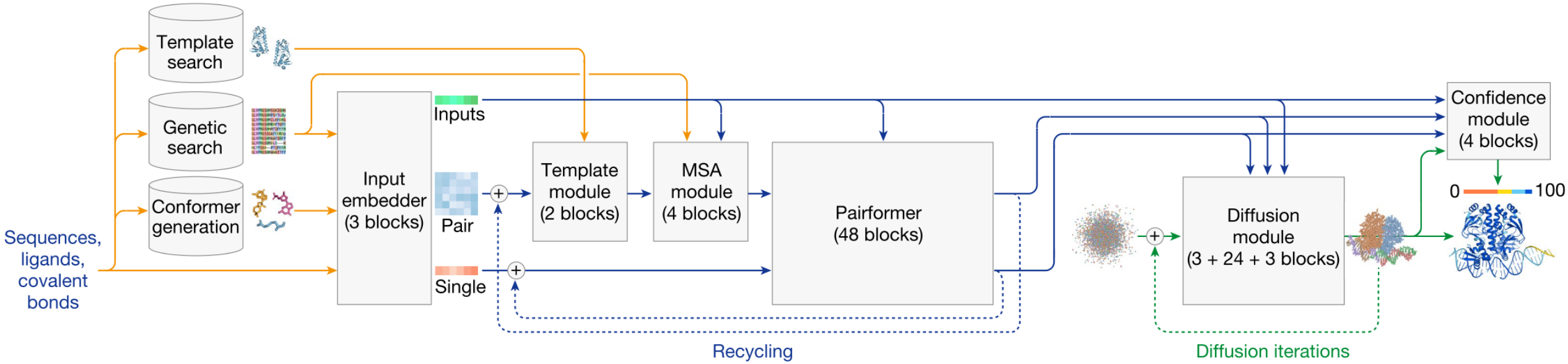
AF3: trained to receive "noised" atomic coordinates and predict correct coordinates



https://www.ebi.ac.uk/training/online/courses/alphafold/

Continuous Evolution: AlphaFold3

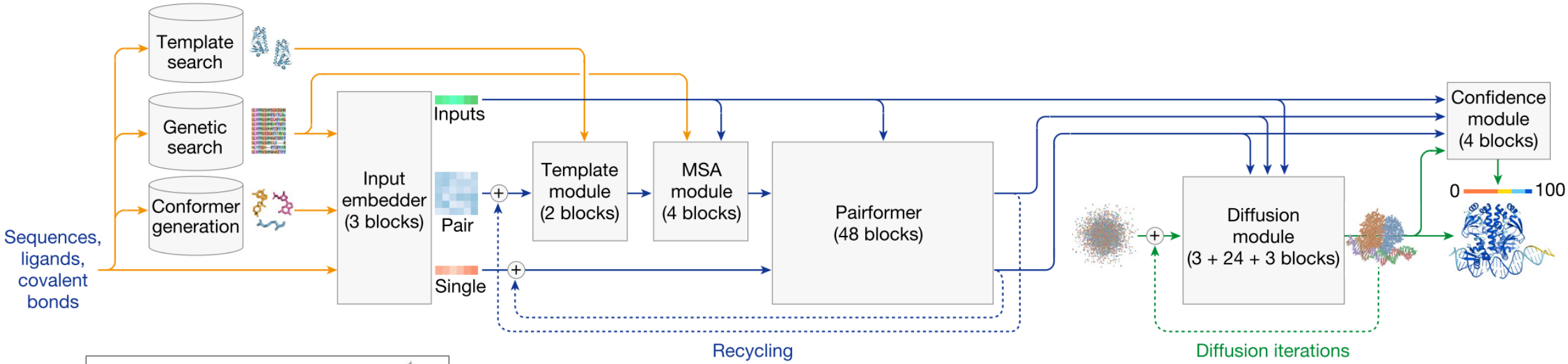
Going beyond Protein Structures to Biomolecules

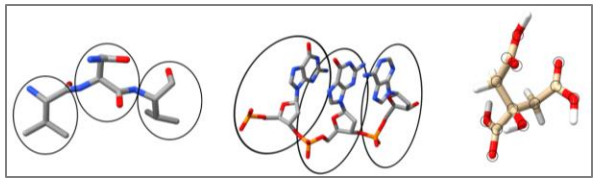


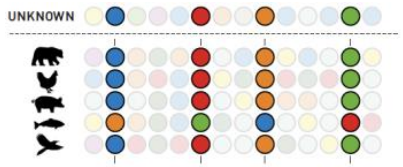
Abramson, J., Adler, J., Dunger, J., … Jumper, J. M. (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, *630*(8016), 493–500. https://doi.org/10.1038/s41586-024-07487-w

Continuous Evolution: AlphaFold3

Going beyond Protein Structures to Biomolecules











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Continuous Evolution: AlphaFold3

Interpreting the Results: the Confidence Module

The predicted structure is accompanied by reliable measures of confidence:

- **pTM:** predicted template modelling

pTM > 0.5 predicted fold similar to the true structure

pTM < 0.5 predicted structure likely wrong

Chain	pTM
0 Leukocidin E	0.88
1 Antibody Fab Light Chain	0.83
2 Antibody Fab Heavy Chain	0.79
3 Centyrin S26	0.75
Full structure	**0.54**



7T82
Leukocidin E
+Antibody Fab
+ Centyrin S26

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- **ipTM:** accuracy of relative positions of chains

ipTM > 0.8 high quality prediction
ipTM < 0.6 likely failed prediction

Chain	ipTM
0 Leukocidin E	0.24
1 Antibody Fab Light Chain	0.35
2 Antibody Fab Heavy Chain	0.35
3 Centyrin S26	0.22
Full structure	**0.41**



7T82
Leukocidin E
+Antibody Fab
+ Centyrin S26

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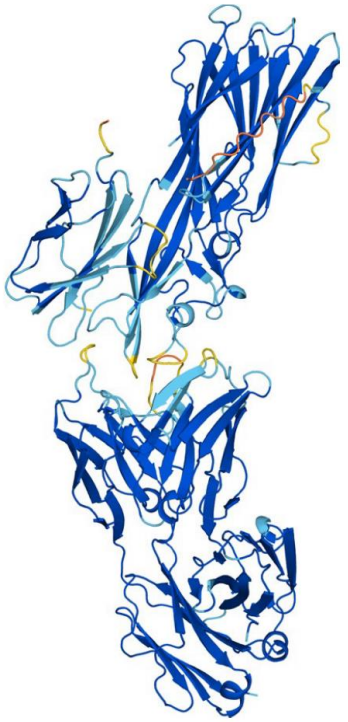
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- **pLDDT:** predicted local distance difference test (based on Cα)



7T82
Leukocidin E
+Antibody Fab
+ Centyrin S26



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Continuous Evolution: AlphaFold3

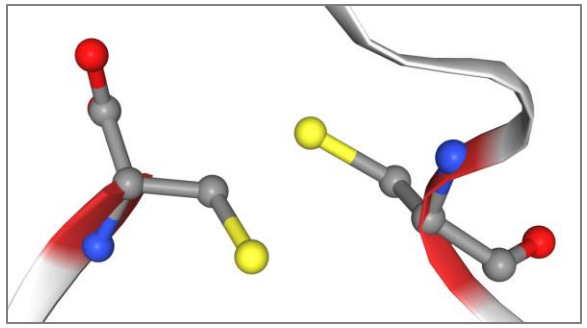
Interpreting the Results: Beyond Confidences

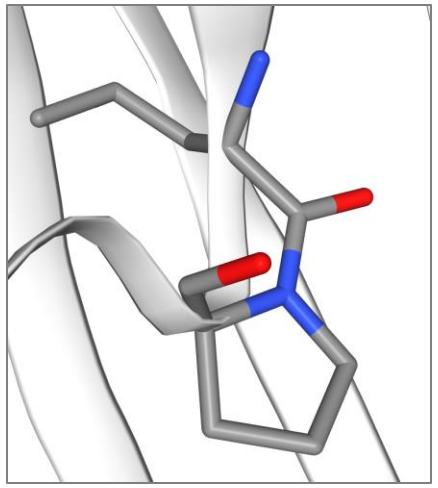
Orthogonal Testing:

- **Structure Assessment**

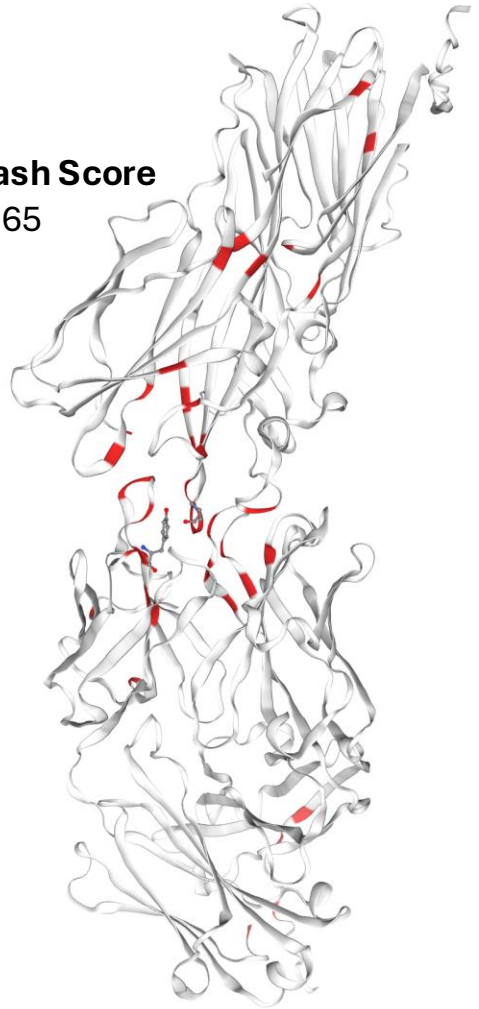
https://swissmodel.expasy.org/assess

  - Stereochemistry: Clashing bonds / atoms
  - Comparison to reference structures
  - Model quality estimation





Clash Score
13.65



Waterhouse AM, Studer G, Robin X, Bienert S, Tauriello G, Schwede T (2024) The structure assessment web server: for proteins, complexes and more. *Nucleic Acids Research*, vol. 52, W1 3863480210.1093/nar/gkae270

How to use the AlphaFold3?

Local & Webserver Solutions

Option 1: https://alphafoldserver.com/
Easy access web server

Option 2: https://github.com/google-deepmind/alphafold3
Local installation on bwForCluster Helix, accessible via bwVisu

```
# list all modules
module avail

# load AlphaFold3
module load bio/alphafold/3.0.1

# Information, examples etc.
module help bio/alphafold/3.0.1
```